



Batterjee Medical College
For Science and Technology

Transport across the cell membrane

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بِسْمِ اللَّهِ الرَّحْمَنِ
الرَّحِيمِ



وَاتَّقُوا اللَّهَ ۖ وَيُعَلِّمُكُمُ
اللَّهُ ۚ وَاللَّهُ بِكُلِّ شَيْءٍ عَلِيمٌ

Learning outcomes:

By the end of the lecture, the student will be able to:

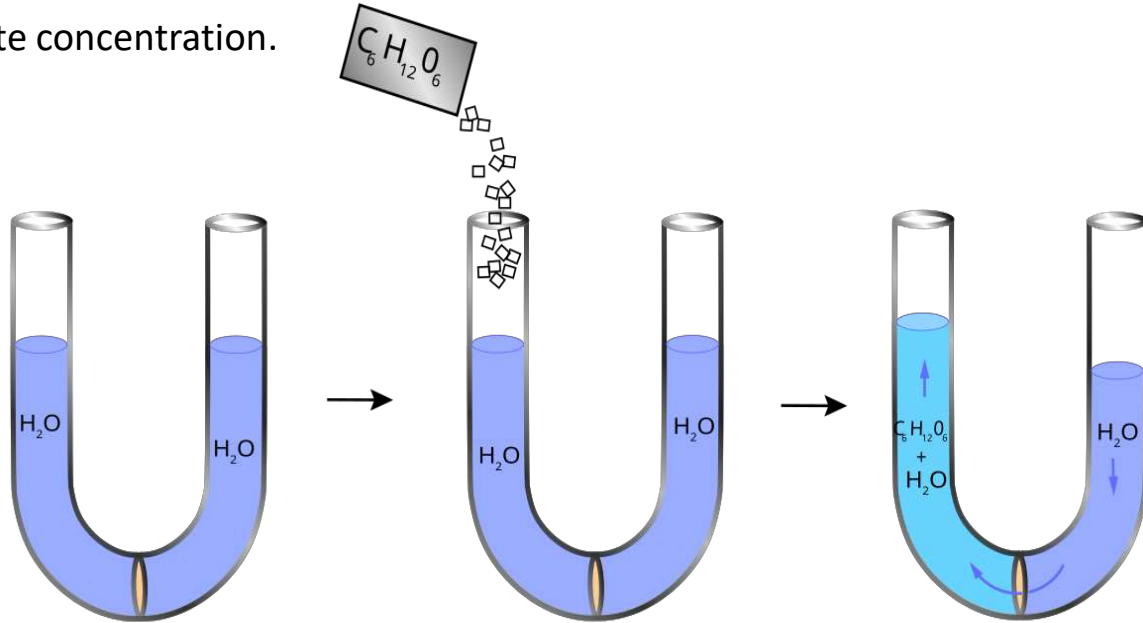
- **Discuss Different methods of transport across the cell membrane**
- **Describe the types of Intercellular communication**

Different methods of transport across the cell membrane

1. Diffusion.

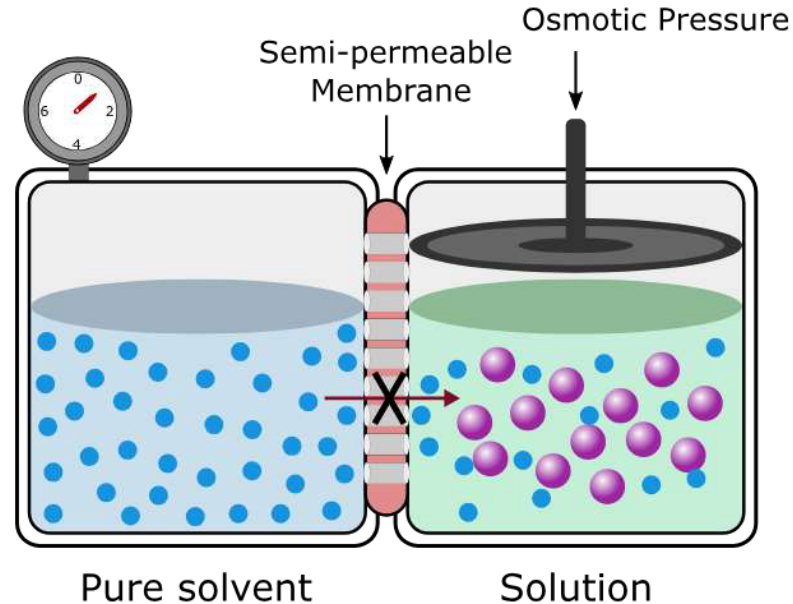
C. Osmosis.

- It is defined as the tendency of solvent particles to diffuse down its concentration gradient, or to an area of higher solute concentration.



Different methods of transport across the cell membrane

- Osmosis (solvent migration) can be prevented by applying pressure to the more concentrated solution, **the osmotic pressure** of the solution.
- The Osmotic pressure is measured in mmHg and is determined by the numbers of particles (molecules or ions) per unit volume of fluid, and not the mass of the particles.
- Osmosis is $\propto$ number of particles/volume of solvent.



Different methods of transport across the cell membrane

- Mole of any substance contains Avogadro's number (**AN**) of atoms = $6.02214076 \times 10^{23}$
- **Osmole** = number of osmotically active particles (molecules) that one mole liberates in solution.
- One mole of glucose = ~~180 gm~~, contains or = one Osmole.
- On the other hand, if the solute dissociates into two ions e.g., NaCl, one mole of NaCl = ~~58.5 gm~~ = two osmoles.
- **Osmolality** = number of osmoles dissolved in one kilogram of water.
- **Osmolarity** = number of osmoles in one liter of water.

Different methods of transport across the cell membrane

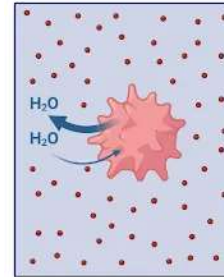
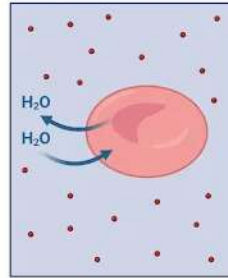
- Plasma osmolarity as regard the sum of all ions (cations and anions) would be over 300 mOsm/L.
- Body fluids are not ideal solutions. Though dissociation of strong electrolytes is complete, the number of particles free to exert an osmotic power is reduced owing to the interaction between ions.
- Osmolarity of plasma is **290 mOsm/L = 5597 mmHg** osmotic pressure.



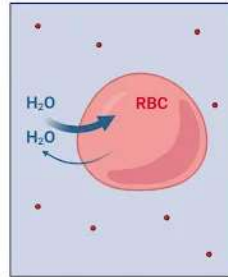
	ICF	ECF	
		Plasma	ISF
Osmolarity (mOsm/L)	290	290	290

Different methods of transport across the cell membrane

- **Tonicity** is the osmolality of a solution relative to the osmolality of plasma.
- An **isotonic solution** has the same osmolality as the plasma i.e. 290 mOsm /L.
e.g. 0.9 % NaCl or 5% glucose solution.
- A **hypertonic solution** has higher osmolality than plasma.
- A **hypotonic solution** has lower osmolality than the plasma.
- All isosmotic solutions would remain so if it were not for the fact that some solutes diffuse into the cells or being metabolized:
- 0.9% NaCl remains isotonic, while 5% glucose is initially isotonic and after being metabolized, the net effect is hypotonic solution infusion.

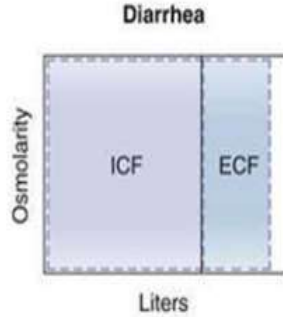


Hypertonic solution

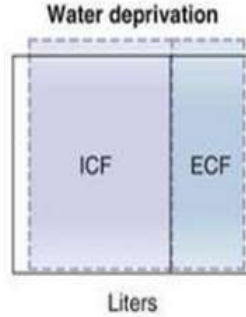
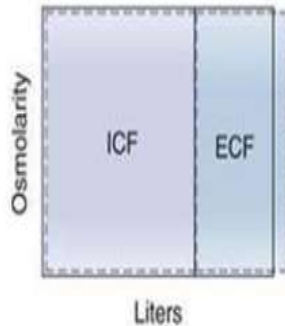


Hypotonic solution

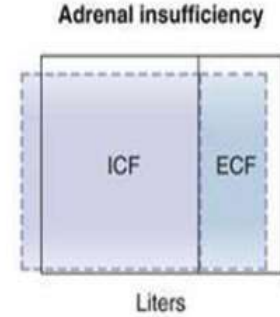
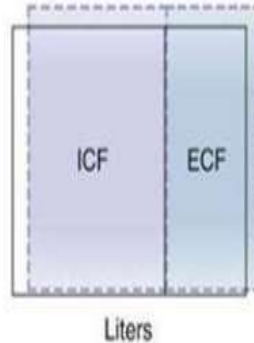
Different methods of transport across the cell membrane



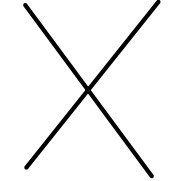
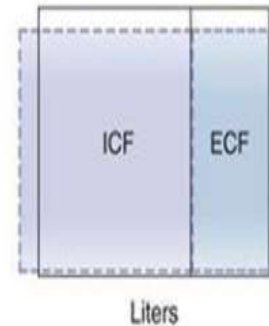
Infusion of
isotonic NaCl



High
NaCl intake



SIADH

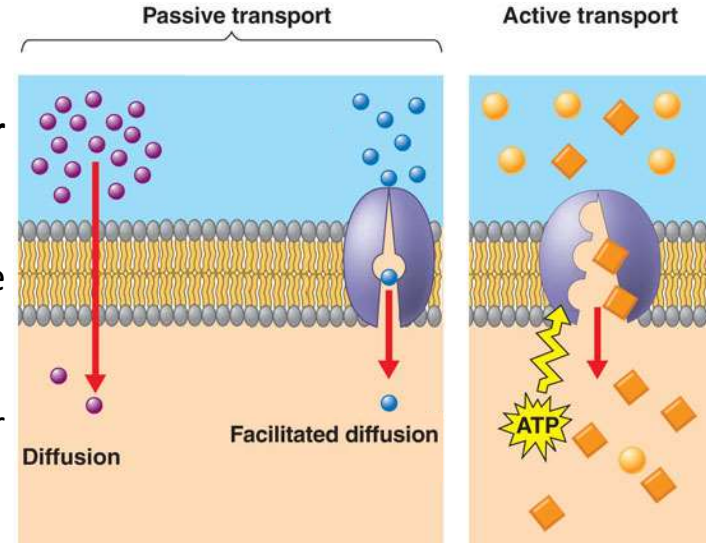


cancel

Different methods of transport across the cell membrane

2. Active transport.

- It is the transport of substance across the cell membrane against an electrochemical gradient.
- Active transport depends on **specific carrier** proteins.
- **Energy** is needed for the substance to be transported.
- The energy is derived from ATP and the carrier protein has ATPase activity.
- The energy required is proportional to $\log \Delta C$.



Different methods of transport across the cell membrane



Blood
+ 0mV

➤ Types of active transport: -

I. **Uniport:** Transports one substance in one direction.
e.g., Ca^{+2} pumps.

II. **Co-transport:** transports two substances simultaneously, which is further subdivided into:

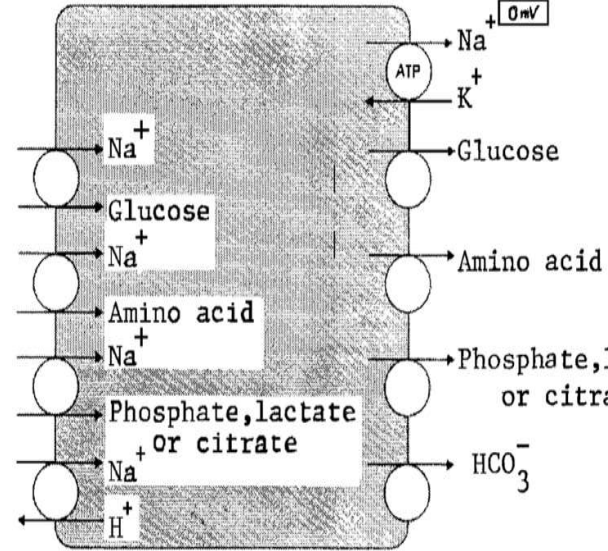
a) **Symport:** Transports two substances in the same direction, e.g., carriers of glucose and Na^+ from the intestinal lumen to inside of the cells.

b) **Antiport:** Transports two substances in two opposite directions, e.g., Na^+/K^+ pump.

➤ Active transport may be classified into:

A. Primary.

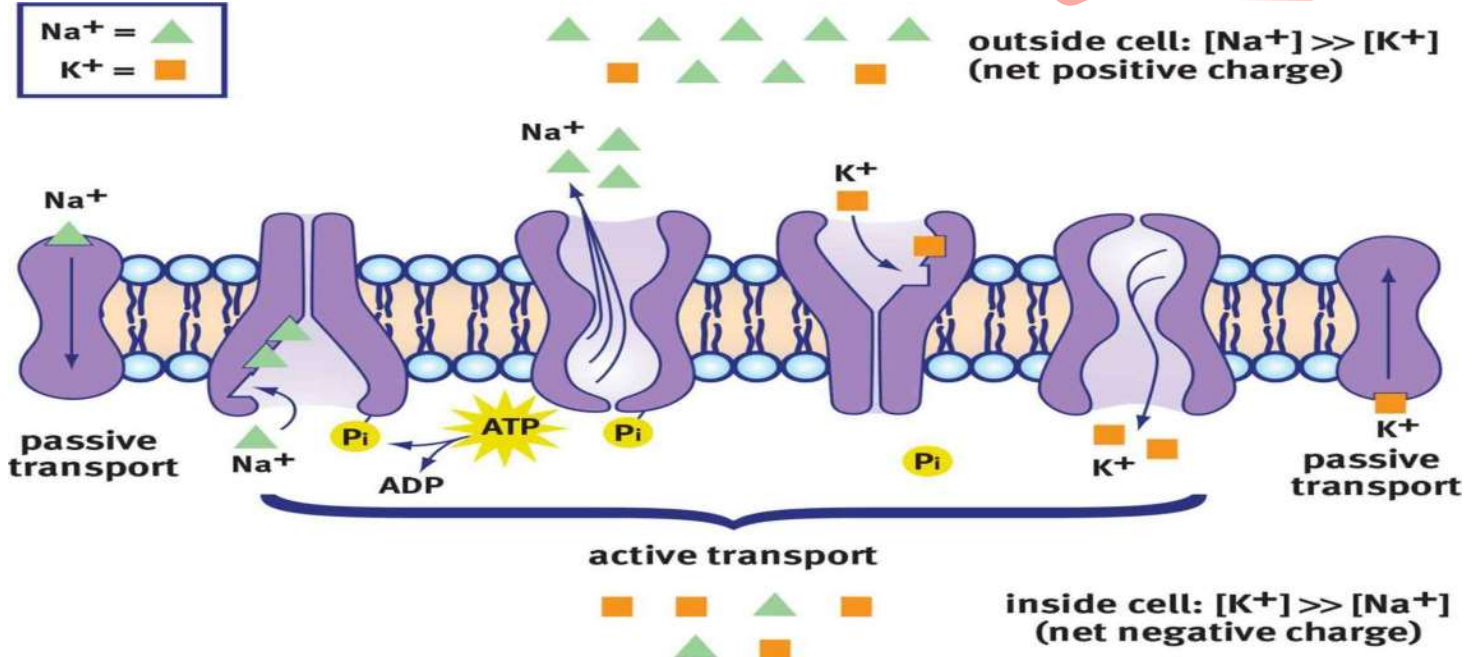
B. Secondary.



Different methods of transport across the cell membrane

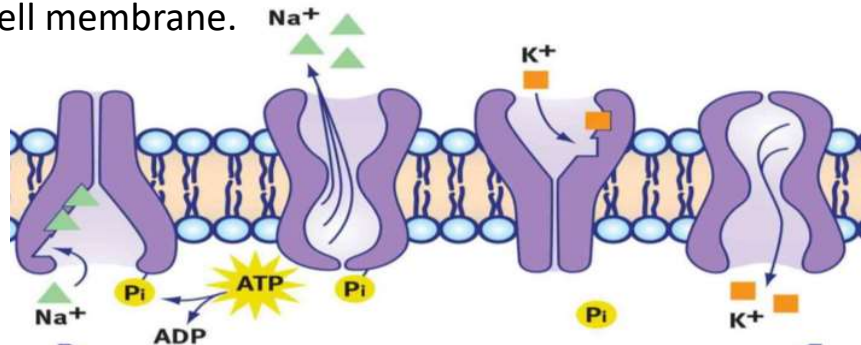
A. Primary (Na^+/K^+ ATPase pump).

- Accounts for 24% of energy utilized by cells, 70% in neurons and 90% in kidneys.



Different methods of transport across the cell membrane

- Has a coupling ratio of 3:2.
- Inhibited by ouabain & related digitalis glycosides.
- Heterodimer, separation eliminates activity:
- The α subunit
 - Have 5 intracellular binding sites for 3 Na^+ , phosphorylation and ATP.
 - Have 3 extracellular binding sites for 2 K^+ and ouabain.
- β subunit: anchor the protein complex in cell membrane.



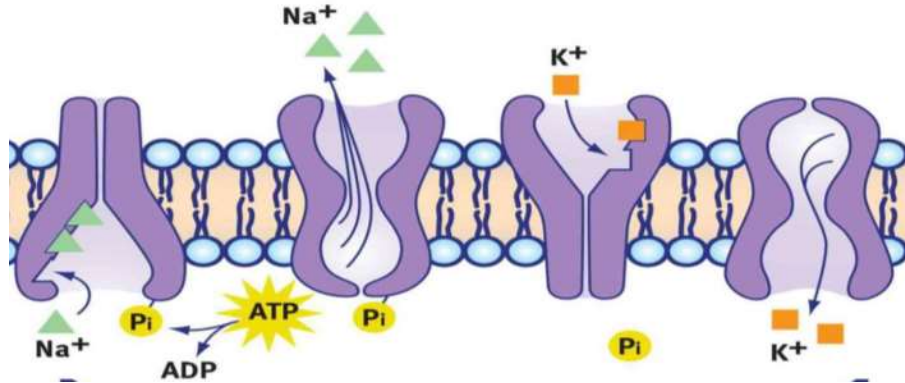
Different methods of transport across the cell membrane

- Binding of 3 Na^+ is followed by ATP binding $\rightarrow$ converted to ADP with P being transferred to phosphorylation site $\rightarrow$ configurationally changes $\rightarrow$ extruding Na^+ extracellularly.
- This is followed by K^+ binding $\rightarrow$ dephosphorylation $\rightarrow$ returning to its original configuration $\rightarrow$ releasing K^+ to the cytoplasm.

- Regulated by:

$\rightarrow \text{NA, K, ATP}$

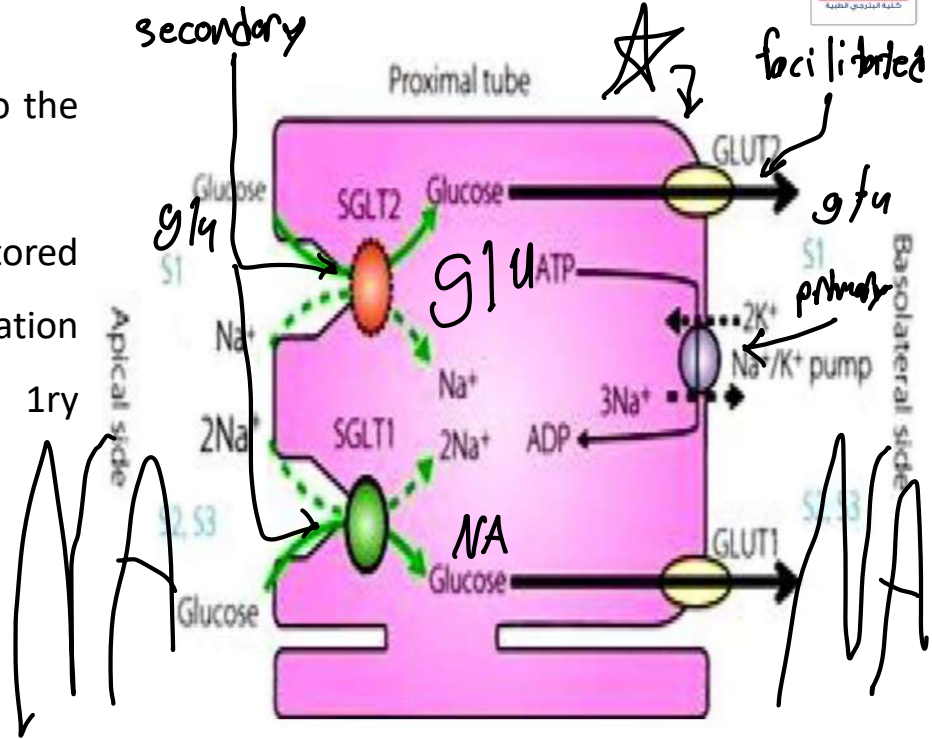
1. Substrate availability.
- ☒ Thyroid hormones (genomic).
 - ☒ Aldosterone (2ry).
 - ☒ Insulin (increases activity by variety of mechanisms).
 - ☒ 2nd messenger molecules (cAMP and DAG).
 - ☒ Dopamine (phosphorylation-mediated inhibition).



Different methods of transport across the cell membrane

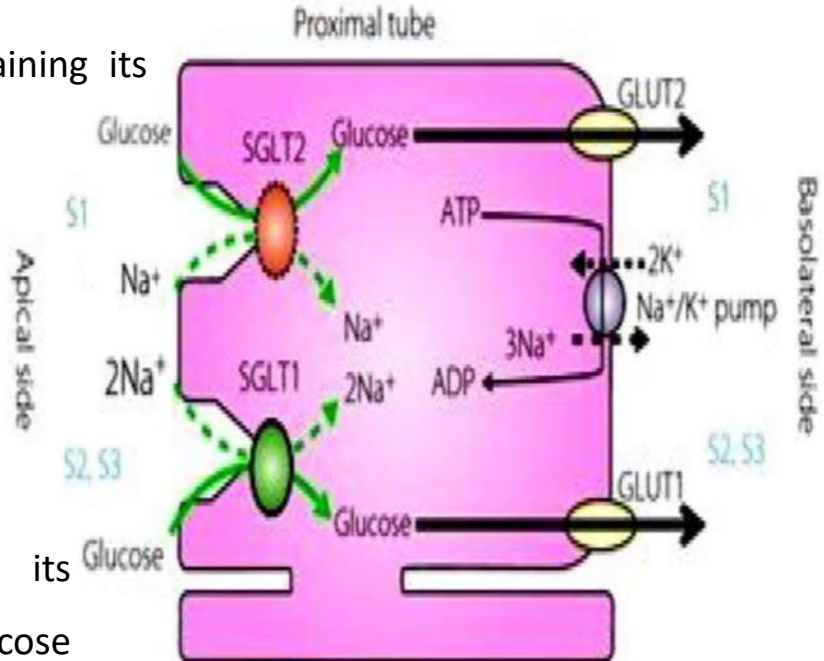
B. Secondary.

- The carriage of glucose secondary to the active transport of Na^+ .
- It gains its energy from the energy stored in the form of ionic concentration differences created originally by primary active transport.



Different methods of transport across the cell membrane

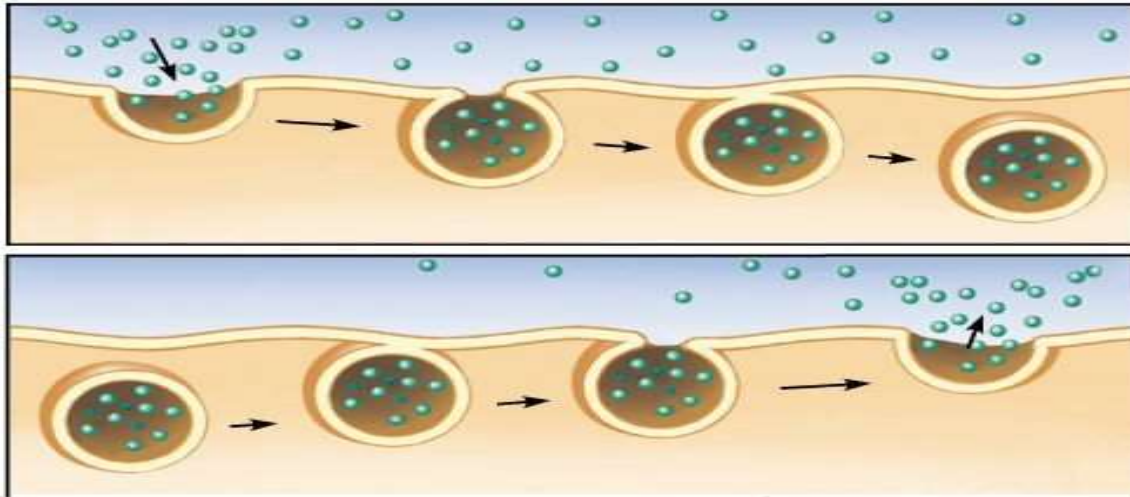
- Na^+ is carried first actively outside the cell.
- This lowers intracellular $[\text{Na}^+]$ → maintaining its chemical gradient.
- Na^+ and glucose bind to the carrier (SGLUT-II or I).
- The carrier changes in shape and so becomes suitable for glucose carriage inside the cell.
- Glucose is then extruded out down its concentration gradient through glucose transporters I & II (facilitated diffusion).



Different methods of transport across the cell membrane

3. Vesicular transport.

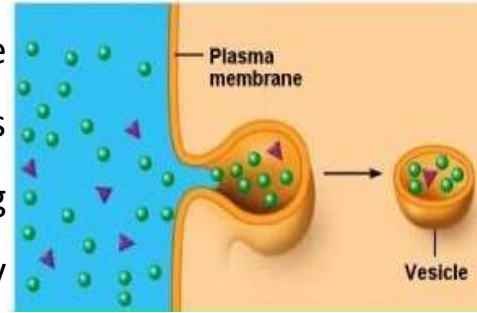
- It is the movement of macromolecules that are enclosed inside cell membrane bound vesicles.
- It is of two types: **endocytosis** and **exocytosis**.



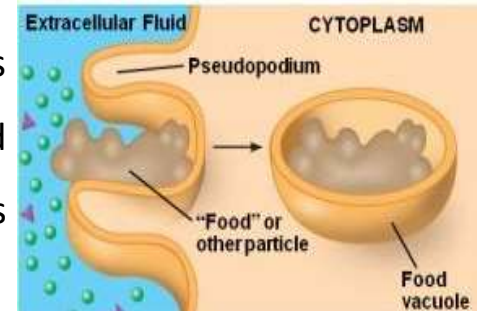
Different methods of transport across the cell membrane

A- Endocytosis: It is the movement from outside the cell to inside.

a) Pinocytosis = cell drinking: It is the only mean by which some very large **water-soluble** macromolecules such as protein molecules can enter the cells as small invagination vesicles containing extracellular fluid. The portion of the membrane then breaks away from the surface of the cell forming a pinocytic vesicle .



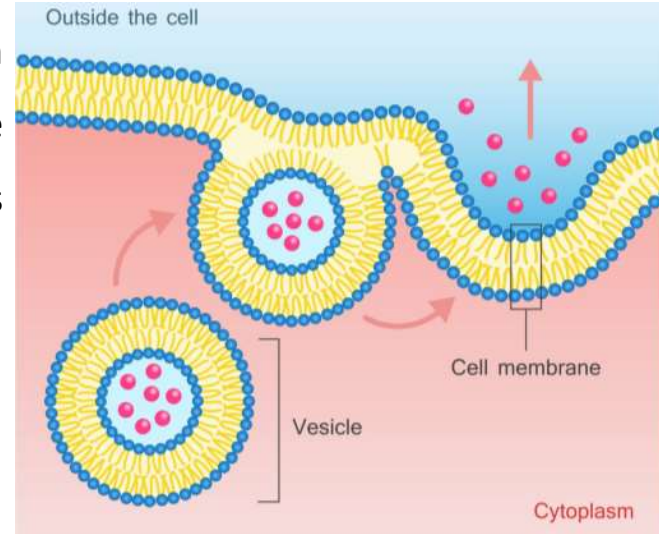
b) Phagocytosis: cell eating: It occurs in the same way as pinocytosis except that it involves **large particulate matter** e.g. bacteria and dead tissue. Only certain cells have the capability of phagocytosis e.g. some white blood cells



Different methods of transport across the cell membrane

B- Exocytosis: It is the movement from inside the cell to the outside.

- **Exo- and endocytosis:** There is a balance between exo- and endocytosis so that the surface area of the cell membrane neither decreases (if endocytosis is more) nor becomes redundant (if exocytosis is more).



Thank You!



Conclusion

Water distributes itself between ICF (40%), ISF (15%) and IVF (plasma, 5%).

Homeostasis is defined as keeping the composition of the internal environment constant.

Cell membrane is an amphipathic structure which contains proteins (55%), phospholipids (25%), cholesterol (13%), other lipids (4%) and carbohydrates (3%).

Conclusion

Transport through the cell membrane may be by diffusion, active transport or vesicular transport.

Intercellular communication may occur through gap junctions, neuronal or endocrine.

References

Guyton and Hall, 13th edition. Unit V(25); 305-14.

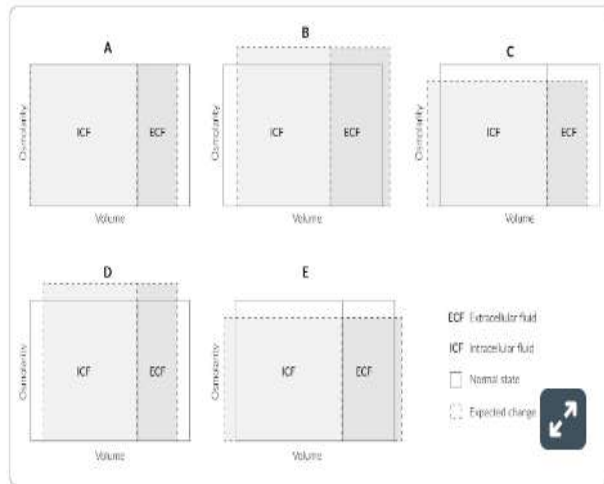
Ganong's review of medical physiology 25th ed. Section I(2); 45-53.

Questions

A 32-year-old woman comes to the physician for increasing fatigue over the past 3 months. During this period, her appetite has decreased and she has lost 10 kg (22 lb). She also reports decreased sexual desire. She has a history of type 1 diabetes. Her pulse is 105/min and blood pressure is 95/63 mm Hg. Physical examination shows darkening of the skin and oral mucous membranes. Serum studies show:

Na ⁺	130 mEq/L
K ⁺	5.2 mEq/L
Glucose (fasting)	74 mg/dL

Which of the following changes to body fluid volume and osmolality would most likely be expected?



Questions



A	A	9%	—
B	B	11%	—
C	C	49%	✓

C shows hyperosmotic fluid loss from the ECF, in which solute loss is greater than volume loss, resulting in **hypovolemic hyponatremia**. This imbalance leads to decreased osmolarity in the ECF, generating a **fluid shift from the ECF to the ICF**, which results in increased ICF volume and decreased ICF osmolarity. This pattern is seen in adrenocortical insufficiency (e.g., Addison disease), in which the deficiency of aldosterone leads to increased excretion of Na^+ . Increased excretion of Na^+ results in higher concentrations of free solute in the urine relative to the ECF (i.e., loss of more solute than free water). Other conditions associated with hypovolemic hyponatremia include excessive use of loop diuretics and 21-hydroxylase deficiency.

 Hyponatremia

 GIVE FEEDBACK

D	D	19%	—
E	E	12%	—

Thank You